host_agent

Where are primary schools in Exeter?

planning_agent

The area is Exeter, so there is 1 entry for Exeter. The entities to search for are primary schools, and we need to find as many entries as possible for primary schools in Exeter.

output steps: ["Find Exeter 1 area", "Search for primary schools in Exeter as many search results"]

named_area

Find the area polygon for Exeter.

Tool call_os_ngd

Arguments: - filters: None - bbox: None - polygon_or_point: True - street_address: None - filename: exeter_area_polygon

Response: Search results have been found. Artifact generated is : exeter_area_polygon, Description is : A geopandas dataframe containing named area data with no filters applied for the query (so remember no search is performed here so spurious entities present) Find the area polygon for Exeter.. count of records fetched : 98.

data_analysis_agent

Filter the artifact 'exeter_area_polygon' to find the specific area polygon for Exeter. Save the filtered results as 'filtered_exeter_area_polygon'.

Tool generate_metadata_for_artifacts

Arguments: - artifact_names: ['exeter_area_polygon']

Response: Metadata generated for 'exeter_area_polygon'.

Tool Code Executor

```python
import matplotlib
matplotlib.use('Agg')
def filter_exeter_area_polygon(data: list):
    gdf = data[0]
    # Filter for Exeter using multiple relevant columns for accuracy
    filtered = gdf[
        (
            (gdf['name1_text'].str.lower() == 'exeter') |
            (gdf['name1_text'].str.lower().str.contains('exeter'))
        ) & (
            (gdf['descriptiongroup'].str.lower().isin(['settlement', 'part of settlement'])) |
            (gdf['description'].str.lower().isin(['city', 'town', 'suburban area']))
        )
    ]
    summary = f"Filtered to {len(filtered)} polygon(s) for Exeter area."
    artifact_name = 'filtered_exeter_area_polygon'
    artifact_description = 'GeoDataFrame containing only the polygon(s) representing the Exeter area, filtered using name and settlement/city/town/suburban are
    return [summary, artifact_name, artifact_description, filtered]
```

Filtered to 1 polygon for the Exeter area. The result is saved as 'filtered_exeter_area_polygon', which contains only the polygon(s) representing the Exeter area, filtered using both the name and relevant settlement/city/town/suburban area attributes.

named_area

The specific area polygon for Exeter has been filtered and saved as 'filtered_exeter_area_polygon'. This file contains only the polygon(s) representing the Exeter area.

land_use_features

Search for primary schools within the area defined by the artifact 'filtered_exeter_area_polygon'.

Tool call_os_ngd

Arguments: - filters: ['School'] - bbox: 'filtered_exeter_area_polygon' - polygon_or_point: True - street_address: False - filename: 'primary_schools_in_exeter'

Response: Multiple search results have been found in multiple datasets. A summary of each is provided. Artifacts generated are : ['landuse_primary_schools_in_exeter'], Descriptions are : ["A geopandas dataframe containing land use data with filters and bbox applied for the query Search for primary schools within the area defined by the artifact 'filtered_exeter_area_polygon'. using filters ['School']. Further Name filtering is available for this"], counts of records fetched are : [45].

data_analysis_agent

Please filter the dataset 'landuse_primary_schools_in_exeter' to specifically identify primary schools. Save the filtered results as 'filtered_primary_schools_exeter'. Include filtering using the 'oslandusetierb' column.

Tool generate_metadata_for_artifacts

Arguments: - artifact_names: ['landuse_primary_schools_in_exeter']

Response: Metadata generated for 'landuse_primary_schools_in_exeter'.

Tool Code Executor

```
import matplotlib
matplotlib.use('Agg')
def filter_primary_schools_exeter(data: list):
    gdf = data[0]
    # Define all oslandusetierb values that indicate a primary school
    primary_school_categories = [
        'Primary School',
        'Primary School,Secondary School',
        'Non State Primary Or Preparatory School',
        'Junior School',
        'Infant School'
    ]
    # Filter for primary schools using oslandusetierb
    filtered = gdf[gdf['oslandusetierb'].isin(primary_school_categories)]
    summary = f"Filtered to {len(filtered)} primary school records in Exeter using oslandusetierb."
    artifact_name = 'filtered_primary_schools_exeter'
    artifact_description = (
        "A GeoDataFrame containing only the primary schools in Exeter, "
        "filtered using the 'oslandusetierb' column. Categories included: "
        f"{', '.join(primary_school_categories)}."
    )
    return [summary, artifact_name, artifact_description, filtered]
```

Filtered to 30 primary school records in Exeter using the 'oslandusetierb' column.

An artifact named 'filtered_primary_schools_exeter' has been generated. This artifact is a GeoDataFrame containing only the primary schools in Exeter, filtered using the 'oslandusetierb' column. The following categories were included: Primary School, Primary School,Secondary School, Non State Primary Or Preparatory School, Junior School, and Infant School.

land_use_features

I found and filtered the primary schools within the area defined by 'filtered_exeter_area_polygon'. The filtered data is saved as 'filtered_primary_schools_exeter'. This dataset contains 30 primary school records in Exeter, filtered using the 'oslandusetierb' column.

Tool generate_metadata_for_all_artifacts

Arguments: - none

Response: Metadata generated for all artifacts.

plotting_agent

Plot the locations of primary schools in Exeter using the artifact 'filtered_primary_schools_exeter' and show the boundary of Exeter from 'filtered_exeter_area_polygon'.

Tool generate_metadata_for_artifacts

Arguments: - artifact_names: ['filtered_primary_schools_exeter', 'filtered_exeter_area_polygon']

Response: Metadata generated for 'filtered_primary_schools_exeter' and 'filtered_exeter_area_polygon'.

Tool Code Executor

```
import matplotlib
matplotlib.use('Agg')
import folium
import geopandas as gpd
from shapely.geometry import Polygon, MultiPolygon

def plot_primary_schools_exeter(data: list):
    # Unpack data
```

```python
    schools_gdf = data[0].copy()
    exeter_gdf = data[1].copy()

    # Ensure CRS is EPSG:4326 for folium
    if schools_gdf.crs is not None and schools_gdf.crs.to_epsg() != 4326:
        schools_gdf = schools_gdf.to_crs(epsg=4326)
    if exeter_gdf.crs is not None and exeter_gdf.crs.to_epsg() != 4326:
        exeter_gdf = exeter_gdf.to_crs(epsg=4326)

    # Remove non-serializable columns for folium plotting
    def drop_nonserializable(gdf):
        # Only keep columns that are object, float, int, or geometry
        keep_types = ['object', 'float64', 'int64', 'geometry']
        cols = [col for col in gdf.columns if str(gdf[col].dtype) in keep_types or col == 'geometry']
        return gdf[cols]
    schools_gdf = drop_nonserializable(schools_gdf)
    exeter_gdf = drop_nonserializable(exeter_gdf)

    # Get centroid of Exeter for map centering
    exeter_centroid = exeter_gdf.geometry.centroid.iloc[0]
    m = folium.Map(location=[exeter_centroid.y, exeter_centroid.x], zoom_start=12, tiles='cartodbpositron')

    # Plot Exeter boundary
    folium.GeoJson(
        exeter_gdf,
        name='Exeter Boundary',
        style_function=lambda x: {'color': 'blue', 'weight': 3, 'fillOpacity': 0.05}
    ).add_to(m)

    # Plot primary schools
    for idx, row in schools_gdf.iterrows():
        popup_text = row['name1_text'] if 'name1_text' in row and row['name1_text'] else 'Primary School'
        folium.GeoJson(
            row['geometry'],
            name=popup_text,
            style_function=lambda x: {'color': 'red', 'weight': 2, 'fillOpacity': 0.3},
            tooltip=popup_text
        ).add_to(m)

    # Save map
    map_filename = 'primary_schools_exeter_map.html'
    m.save(map_filename)

    # Prepare summary
    first5 = schools_gdf[['name1_text', 'oslandusetierb', 'geometry_area_m2']].head(5)
    summary = f"Plotted {len(schools_gdf)} primary school locations in Exeter. First 5 schools: {first5.to_dict(orient='records')}"
    description = "A folium map showing the locations of all primary schools in Exeter (as polygons), overlaid with the Exeter city boundary. Each school is sh
    return [summary, 'primary_schools_exeter_map.html', description, map_filename]
```

```
# The function is defined as required by the template.
```

Plotted 30 primary school locations in Exeter. First 5 schools: [{'name1_text': 'Trinity School Church of England Primary and Nursery School', 'oslandusetierb': 'Primary School', 'geometry_area_m2': 17633.04}, {'name1_text': 'The Stansfield Academy', 'oslandusetierb': 'Primary School,Secondary School', 'geometry_area_m2': 4943.501}, {'name1_text': 'Clyst Heath Nursery and Community Primary School', 'oslandusetierb': 'Primary School', 'geometry_area_m2': 14304.149}, {'name1_text': 'St Nicholas Roman Catholic Primary School', 'oslandusetierb': 'Primary School', 'geometry_area_m2': 16733.162}, {'name1_text': 'Countess Wear Community School', 'oslandusetierb': 'Primary School', 'geometry_area_m2': 26690.855}] Artifact name: primary_schools_exeter_map.html Description: A folium map showing the locations of all primary schools in Exeter (as polygons), overlaid with the Exeter city boundary. Each school is shown in red, and the city boundary in blue. File: primary_schools_exeter_map.html

host_agent

The locations of primary schools in Exeter have been identified and mapped. Here’s what was done:

- The Exeter city boundary was found and used as the search area.
- All primary schools within Exeter were located using official land use data.
- A map was generated showing the locations of 30 primary schools in Exeter, with each school highlighted and the city boundary shown for reference.

If you would like to view or download the map, let me know! I can also provide a list of the schools or more details about any specific one.